MAHA-GR-ALIGN

Compliance audit against the Software Requirements Specification

AI-Assisted Drafting and Legal Alignment System for Maharashtra Government Resolutions

VJTI AI Hackathon 2026 — Problem Area 1

Audited 2 August 2026 against `Maha-GR-Align_Simple_SRS-updated.pdf`

1 Executive summary

Group	Verdict	Basis
FR-1 -- FR-3	**MET**	Extraction, archive verification and highlighting, live on an uploaded document
FR-4 -- FR-6	**MET**	23,977-clause vector index, cross-department search, score and source link
FR-7 -- FR-9	**MET**	31-term controlled glossary, flagging and suggestion
FR-10 -- FR-12	**MET**	Manual of Office Procedure Appendix 10.6, deterministic
FR-13, FR-15, FR-16	**MET**	Upload and type; audit trail; PDF and DOCX
FR-14	**MET**	Original text, AI suggestion and validation results side by side, with accept or discard
NFR-1, 2, 5, 6	**MET**	10 ms against a 30 s budget; on-premise; Unicode; provenance on every finding
NFR-3, NFR-4	**NOT MET**	Prototype: not yet trialled with a desk officer, not yet deployed
Overall	**All 16 functional requirements met.**	

Every figure in this report was measured on the real corpus and is reproducible by a command named in Section 8.

Beyond the specification, the system implements two checks the SRS does not ask for and which address the problem statement's ``cross-department conflict detection'' more directly than similarity search can:

- **Cross-document contradiction** --- a draft that restates a figure from the order it cites, differently from what that order says, is caught exactly. Demonstrated by two documents that each pass every other check.
- **Blast radius** --- before a draft that cancels an order is issued, the system reports how many GRs still in force depend on that order, and in which departments.

2 Scale

Government Resolutions indexed	17,283
Departments	4
Archive coverage	2017-10-12 to 2026-07-22
Resolved citation edges	4,815
of which cross-departmental	1,538
typed SUPERSEDES / AMENDS	33 / 169
Clauses in the vector index	23,977 (from 6,000 GRs)
searchable after masking procedural boilerplate	21,018
Controlled glossary terms	31
Finding codes across four checkers	13 reference, 15 template, 3 glossary, 1 contradiction
Automated tests, zero network calls	131

3 Functional requirements

3.1 Reference Parser Module

Req	Verdict	Evidence
FR-1	**MET**	`references.py` extracts cited orders from the preamble and the operative body, and classifies nine document kinds: Government Resolution, circular, corrigendum, notification, order, **court order**, Act, letter, minutes. The distinction is load-bearing --- the archive holds departmental orders and never holds court orders, so classifying before judging is what stops a High Court order being reported as a missing GR.
FR-2	**MET**	Each citation resolves against the 17,283-node index on two keys: the Devanagari file number (exact, or a script-independent digit key) *and* the date within a $\pm$3-day window. Neither alone suffices --- 41.1% of GRs share a file number, and a date-only resolver produced 19,840 edges with five GRs falsely tied at in-degree 292.
FR-3	**MET**	Thirteen finding codes with line numbers and click-to-highlight in the document.

Measured extraction: 98.9% of date-bearing reference items (3,461 of 3,501 across 906 GRs), against an acceptance threshold of 80%. Reaching it required supporting spelled-out Marathi dates --- दिनांक ०९ जुलै, २०२६ --- matched on a matra-stripped skeleton so that a visually-reordered PDF text layer still parses correctly.

3.2 Semantic Conflict Detector

Req	Verdict	Evidence
FR-4	**MET**	`embed.py` extracts operative clauses and encodes them with MahaSBERT (`l3cube-pune/marathi-sentence-similarity-sbert`, 768-dim, CPU). Only the operative section is indexed: headers and वाचा blocks are near-identical corpus-wide and would make every GR the nearest neighbour of every other.

Req	Verdict	Evidence
FR-5	**MET**	23,977 clauses from 6,000 GRs; 21,018 searchable after masking procedural boilerplate. Search excludes the drafting officer's own department by default, which is the requirement's point --- within a department the officer already knows the standing orders. Measured live: **0.6 s** per draft.
FR-6	**MET**	Each hit shows the cosine score, a band, the source department and date, and a link that opens the source GR with its supersession status and its official `gr.maharashtra.gov.in` URL.

Retrieval quality. Procedural furniture --- the Finance-concurrence line, the budget-head line --- appears near-verbatim in a large share of all GRs and matched everything before it was masked, at scores of 0.87--0.91. With 2,959 such clauses (12%) excluded, hits are substantive: an infrastructure-funding draft returns other departments' infrastructure-funding clauses at 0.66--0.69, and a fund-release draft returns fund-release clauses at 0.71--0.76. The masking is applied at *search* time rather than index time, so the exclusion list can be tightened without re-encoding.

Similarity is reported as a score against a stated threshold rather than as a verdict, and every hit is presented as a candidate for the officer to read. This matches FR-6's wording --- the requirement asks for *potential* conflicts.

3.3 Bilingual Terminology Module

Req	Verdict	Evidence
FR-7	**MET**	`data/glossary.json`: 31 terms, each citing its authority --- the Manual of Office Procedure or SRS FR-11. `make glossary` re-measures corpus evidence only and never overwrites a curated decision.
FR-8	**MET**	Non-approved renderings are flagged with line numbers and occurrence counts.
FR-9	**MET**	The approved term is suggested, *except* where a replacement cannot be justified from one side of the document: रद्द (cancelled), अधिक्रमित (superseded) and व्यपगत (lapsed) do different things to an order, so ``cancelled'' in an English GR may be a correct rendering of रद्द. Those report the ambiguity and withhold the suggestion.

The finding this module exists for. Across 2,500 corpus documents, ``Government Decision'' appears in **1,861** and ``Government Resolution'' in **1,021** --- and the Manual of Office Procedure uses ``Government Resolution'' six times and ``Government Decision'' **not once**. The dominant machine-translated rendering is the non-standard one. No amount of mining the corpus could reveal this, which is exactly why FR-7 asks for a *controlled* list rather than a discovered one.

3.4 Template Enforcement Engine

Req	Verdict	Evidence
FR-10	**MET**	Structure declared in `data/gr_template.json` from Manual Appendix 10.6. No model call, and a test enforces that.

Req	Verdict	Evidence
FR-11	**MET**	All four named items verified: preamble ordering (Background → Reference → Resolution) as codes T20--T22; financial sanction block; signature headers; and the budget head, matched on digits rather than separators --- a hyphen-anchored regex finds **zero** budget heads in 1,500 real GRs, because the corpus writes them space-separated and in Devanagari digits. `2054-00-101-02` and २२०३-००-८००-०१ both match.
FR-12	**MET**	Errors, warnings and advisories, each citing the Manual clause that triggered it.

Two rule sets, one engine. A document *we generate* is held to the full Appendix 10.6 form. An *already-issued* GR read back through OCR is scored differently, because sections that are images in the signed PDF cannot be read by any text pipeline --- the signature block appears in 30.4% of the corpus and the Governor's order line in 2.2%. Under the drafting rules 0.2% of 400 real GRs score compliant, which measures the scan rather than the drafting; under the issued rules, 45.0%. Every section not counted is named in the response, so nothing is silently forgiven.

3.5 User Interface and Workflow

Req	Verdict	Evidence
FR-13	**MET**	Five pages: Home, Draft GR (type), Verify GR (upload PDF or scan), Check Graph, Recent GRs (upload to a pre-issue in-tray). Extraction tries the PDF text layer first and OCRs only scans, reporting which was used --- OCR over a document you can already read exactly introduces errors rather than removing them.
FR-14	**MET**	A refinement is *held as a proposal* rather than written to the preview, so the original and the suggestion exist at the same moment. The two are shown in aligned columns with a line-level diff, above a before-and-after compliance count --- the requirement's three elements in one view. Nothing is written until the officer accepts; discarding leaves the document untouched. Validation results also appear on tabbed panels beside the document: Template, Citations and Terminology on Verify; Template and Other Departments on Draft.
FR-15	**MET**	Version history tags every entry Template / AI / Officer with a timestamp and a restore button. Hand edits become versions once typing settles.
FR-16	**MET**	PDF and DOCX generated from one document model, so the two cannot drift.

4 Engineering guardrails

The system is designed so that a wrong answer is structurally hard to produce. Each guardrail below is enforced in code, and most are pinned by a test.

Facts are computed, never generated

No model call exists in `template.py`, `references.py`, `conflicts.py` or `glossary.py`, and a test asserts the absence of the imports that would allow one. Whether a draft carries a signature block, whether a citation

resolves, whether two figures differ --- each is settled by a regex or an integer comparison. A language model may *explain* a finding; it is never the reason the system believes one.

The alignment verdict is the clearest case. Given the findings and asked to judge, the local model concluded that a document ``remains valid because the cited references are either unverified or superseded'' --- about a document whose citations had in fact been superseded. The verdict is therefore computed from the dependency graph and handed to the model to phrase.

Model output is filtered before it reaches the officer

- **Identifiers are withheld from the prompt.** The narrative reviewer is never shown an order number, date or line number; citations are identified to it only as `[1]`, `[2]`. It cannot corrupt an identifier it never saw.
- **Any sentence containing an unexpected digit is discarded whole**, not repaired. A mangled file number reads exactly as authoritative as a correct one, so the only safe action is to drop the sentence and report how many were dropped.
- **Summaries are gated on figures.** Every monetary amount in the document is extracted deterministically and whitelisted, including its crore and lakh readings; a sentence quoting any other figure is removed. A summary with a wrong rupee figure is worse than no summary.

The five ways of not finding something are never merged

A citation that does not resolve is reported under one of five distinct verdicts: `not_in_archive` (the only one that is a defect in the document), `outside_range`, `other_dept`, `out_of_scope` and `unverifiable`. Collapsing them into ``missing'' would roughly quadruple the error count and make it false --- on one sample GR, 5 genuine misses against 12 citations that simply predate the archive. Court orders and Acts are reported as outside what a GR archive holds, never as missing references.

Abstention is a first-class outcome

Ambiguity abstains rather than guessing. If a file number matches a series but no member falls in the date window, the citation is left unresolved rather than snapped to the nearest sibling. If a citation gives a date but no file number, no resolution is offered --- a date alone cannot identify an order, because every GR issued that day would match. If a glossary term's correct replacement cannot be established from one side of a bilingual document, the suggestion is withheld and the ambiguity is stated.

Evidence travels with every claim

Every graph edge carries the verbatim Marathi sentence that produced it, and that sentence is shown wherever the edge is used to justify a finding. Template findings cite the Manual clause. Glossary findings cite the Manual or state that the entry is a proposal. Semantic hits show the source GR, its department, its date and a link to the signed PDF on the Government's own site. An edge whose justification cannot be displayed does not ship.

Degradation is visible, never silent

If the local model is unreachable, the affected panel says so and shows the computed result instead. If the corpus is absent, the pages that need it grey out rather than failing on click. If OCR produced the text rather than a PDF text

layer, the interface says which and shows the confidence, because those warrant different trust. ``The model was unavailable'' is a legitimate state to display; a fabricated answer is not.

5 Statistics

Accuracy and coverage	
Citation extraction, date-bearing reference items	98.9%
Resolver precision, stratified sample	48/50
digit-key matches	14/14
cross-department	12/12
समक्रमांक inherited number	6/6
exact number	10/10
date-drifted	6/8
Edges with an exact date match	88.6%
Template compliance, issued rule set, 400 GRs	45.0%
Performance	
All four deterministic checks, one GR	10 ms
Citation check alone	4 ms
Semantic search, warm	0.6 s
Corpus screening, 15 GRs	170 ms
NFR-1 budget	30,000 ms
Findings across the corpus	
Dead references (GRs in force citing superseded orders)	24
rotted / born dead / latent cross-department	10 / 14 / 2
Largest blast radius, single Finance circular	229 live dependents
Bilingual magnitude errors detected	335
Inconsistently rendered legal terms	26
``Government Decision'' vs ``Government Resolution''	1,861 vs 1,021

6 Architecture (SRS §5)

Layer	As built
1. Data	mahGRs corpus, PDF→text→embeddings, with an EasyOCR (Marathi + English) fallback for scans.
2. NLP engine	MahaSBERT for Marathi sentence embeddings, IndicTrans2 for translation, Qwen3.5-2B for generation --- all running on-premise, on CPU.
3. Vector search	Exact NumPy inner product over a $23{,}977 \times 768$ matrix, a few milliseconds per query. Chosen over FAISS (``or similar'' in the SRS) because at this scale the exact scores are cheap, and an SDC deployment needs one fewer dependency.
4. Compliance rules	`data/gr_template.json` (Manual Appendix 10.6) and `data/glossary.json` (controlled terminology), both plain JSON and editable without code changes.
5. Frontend	React + Vite dashboard, five pages, bilingual interface, vendored Devanagari fonts so it renders correctly with no network.

7 Acceptance criteria (SRS §9)

#	Criterion	Verdict	Measured
1	Extracts $\geq$80% of cited references from a sample preamble	**MET**	**98.9%** --- 3,461 of 3,501 date-bearing reference items across 906 GRs
2	Two conflicting clauses flagged with a similarity score above a threshold	**MET**	Cross-department hits scored 0.66--0.76 above a 0.55 threshold; separately, a Rs. 5,00,00,000 contradiction detected exactly between two otherwise-clean documents
3	A generated draft contains all mandatory template sections	**MET**	Both demonstration GRs compliant in draft mode --- all 15 sections including budget head, Finance concurrence, Governor's order line and signature
4	Marathi--English term mismatches highlighted against a small glossary	**MET**	31-term glossary; ``Government Decision'' flagged against the Manual's ``Government Resolution'' with a suggested replacement
5	A working web prototype allows upload and viewing validation results	**MET**	Five pages, three upload paths, all checks live

8 Reproducing every figure

Figure	Command
Graph scale, edge types	`make graph`
Dead references	`make audit`
All reported numbers	`make metrics`
Resolver precision sample	`make precision`
Glossary and corpus evidence	`make glossary`
Clause vector index	`make index`
Citation report for one document	`make refcheck FILE=...`
Test suite (131, no network)	`make test`
Service health	`make doctor`

Corpus: orgpedia/mahGRs (MIT licence). Template authority: Maharashtra Manual of Office Procedures, 2023, Appendix 10.6. Prototype built for the VJTI AI Hackathon 2026; not an official Government of Maharashtra service.